



**HEXAGON
TECH HUB**

2025 **PROJECT ON DATA ANALYSIS**

**TOPIC: “UNDERSTANDING AUTO INSURANCE
CLAIMS: A DEMOGRAPHIC, BEHAVIOURAL,
AND GEOGRAPHIC ANALYSIS”**

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INTRODUCTION

Auto insurance plays a critical role in protecting individuals and businesses from financial losses arising from vehicle-related risks such as accidents, theft, and damage. For insurance companies, the ability to accurately assess risk, price policies appropriately, and manage claims efficiently is essential for sustaining profitability and competitiveness. With the increasing availability of customer and claims data, data analytics has become a vital tool for understanding claim behaviour and policyholder characteristics.

This project analyzes an auto insurance dataset containing anonymized information on policyholders' demographics, vehicle attributes, claim frequencies, and claim amounts. The aim is to identify patterns in insurance claims, assess key risk factors associated with higher claim frequency and severity, and segment customers based on their risk and value profiles. By examining demographic, behavioural, and geographic factors, the analysis provides insights into how different policyholder characteristics influence insurance outcomes.

The findings from this analysis are intended to support data-driven decision-making in areas such as risk assessment, premium optimization, and targeted marketing strategies. Ultimately, the project demonstrates how business intelligence tools such as Microsoft Excel and Power BI can be used to transform raw insurance data into actionable insights that enhance operational efficiency and customer value.

QUESTIONS

Claim Frequency and Amount Analysis

- What are the average claim frequencies and amounts for different demographic groups (e.g., age, gender, marital status)?
- Are there any specific vehicle characteristics (e.g., make, model, year) that correlate with higher claim frequencies or amounts?

Risk Assessment

- Which factors (e.g., household income, education level, coverage zone) are most indicative of high-risk policyholders?
- Can we identify any common characteristics among policyholders who make frequent claims?

Customer Segmentation and Marketing

- What are the key characteristics of policyholders with low claim frequencies and high household incomes?
- How can we segment the customer base to identify high-value customers for targeted marketing campaigns?

Demographic Analysis

- How does the distribution of policyholders vary across different demographic factors (age, gender, marital status)?
- Are there any noticeable trends in car usage and ownership among different demographic groups?

Geographical Analysis

- How do claim frequencies and amounts vary across different coverage zones?
- Are there any regional trends or patterns that should be taken into consideration for marketing or risk assessment?

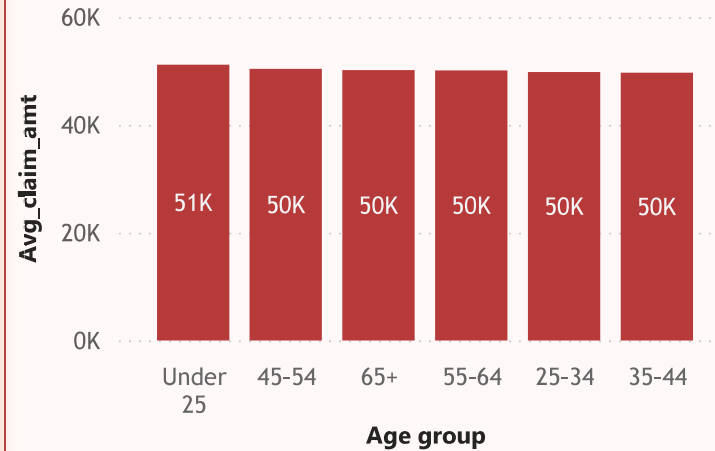
Customer Behavior Insights

- How does the presence of children driving affect the frequency and number of claims?

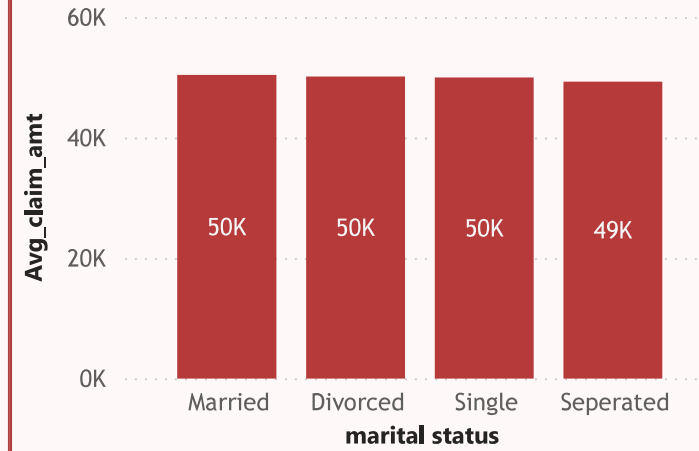
TOTAL CLAIM AMOUNT IN USD

1.88bn

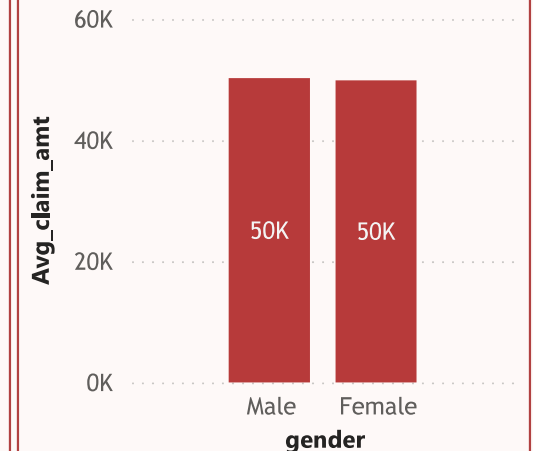
Average Claim Amount of each Age Group



Average Claim Amount of each Marital Status



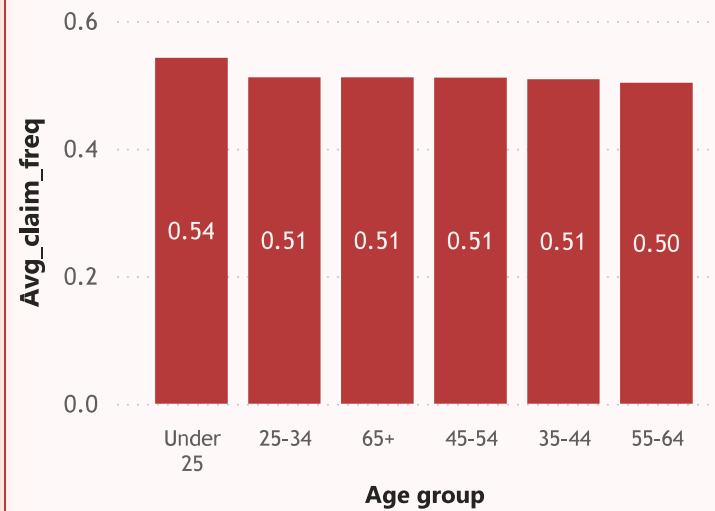
Average Claim Amount by each Gender



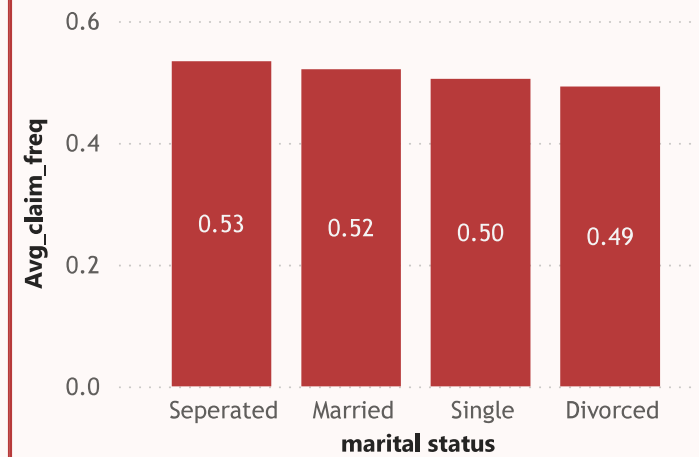
TOTAL CLAIM FREQUENCY

19K

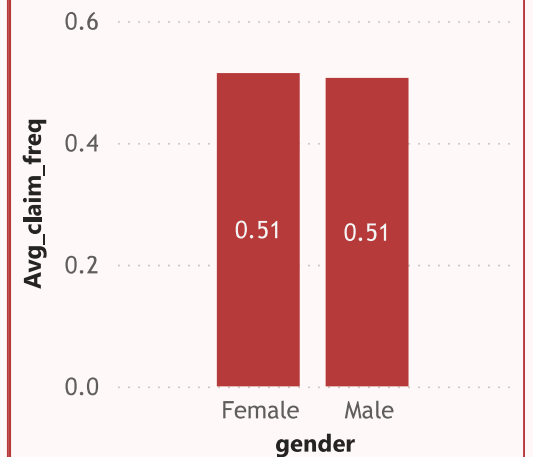
Average claim frequency by Age group



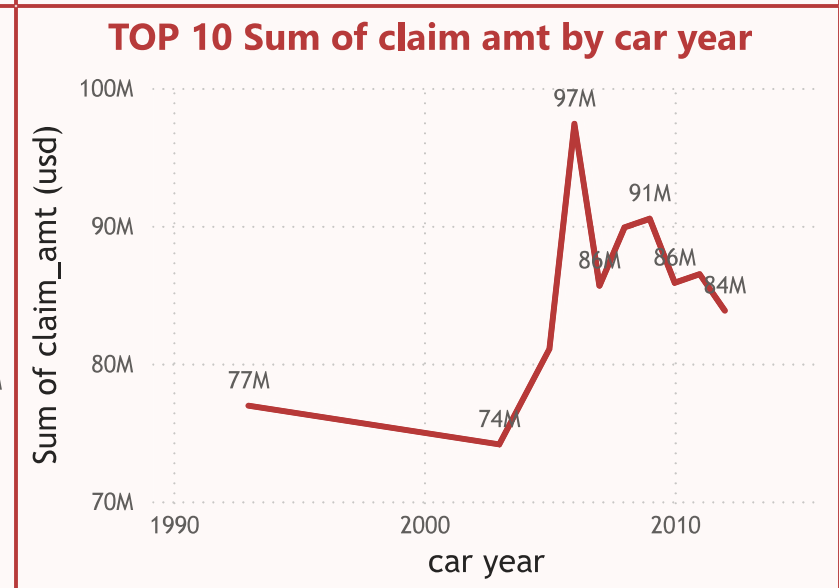
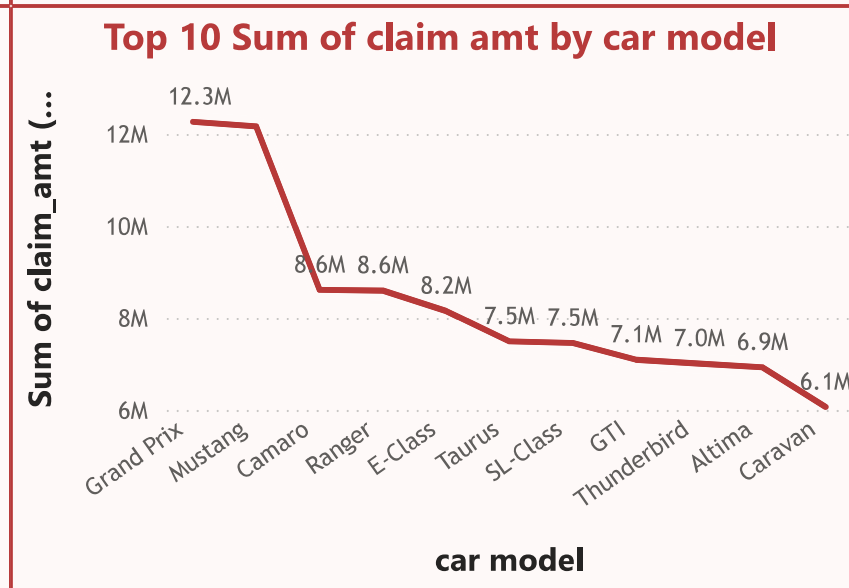
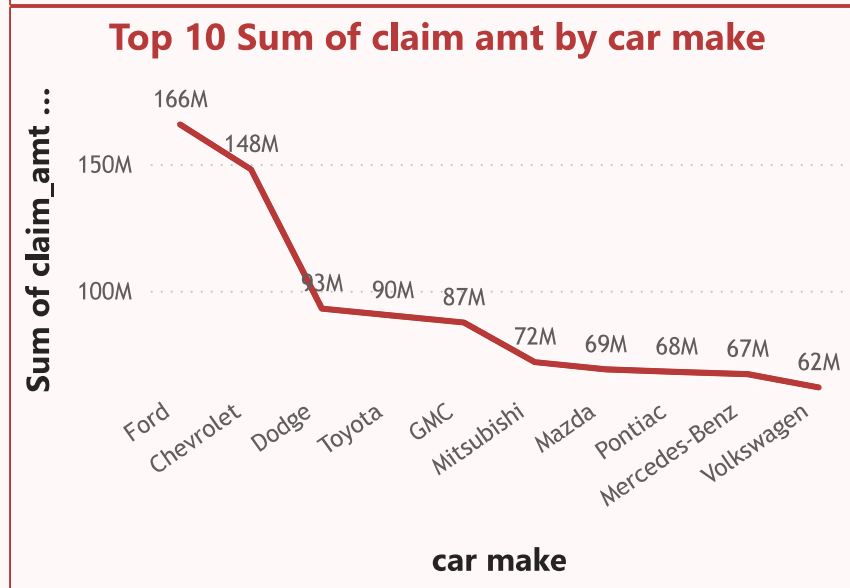
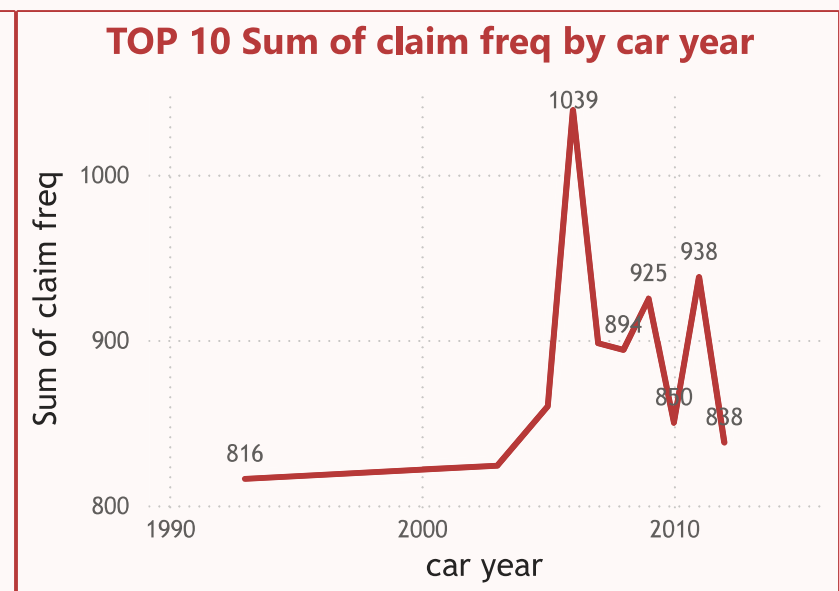
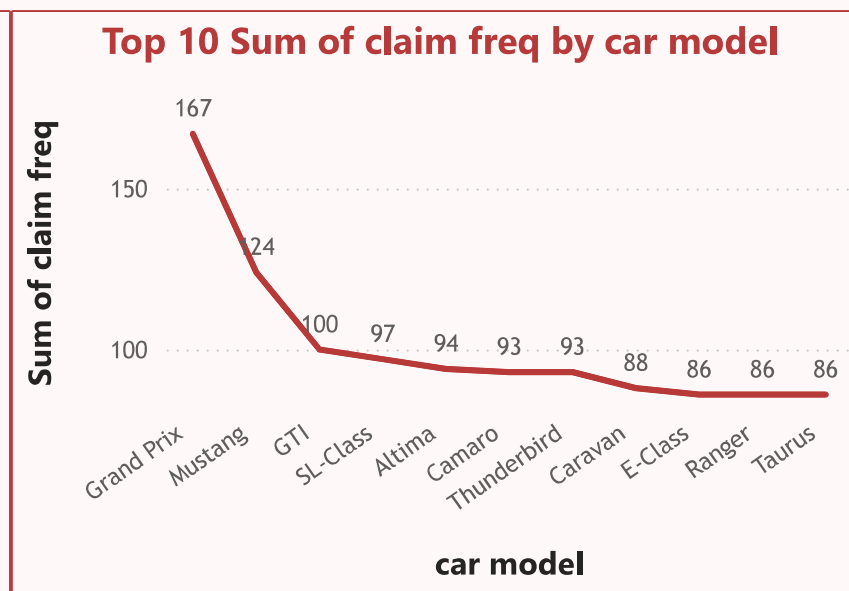
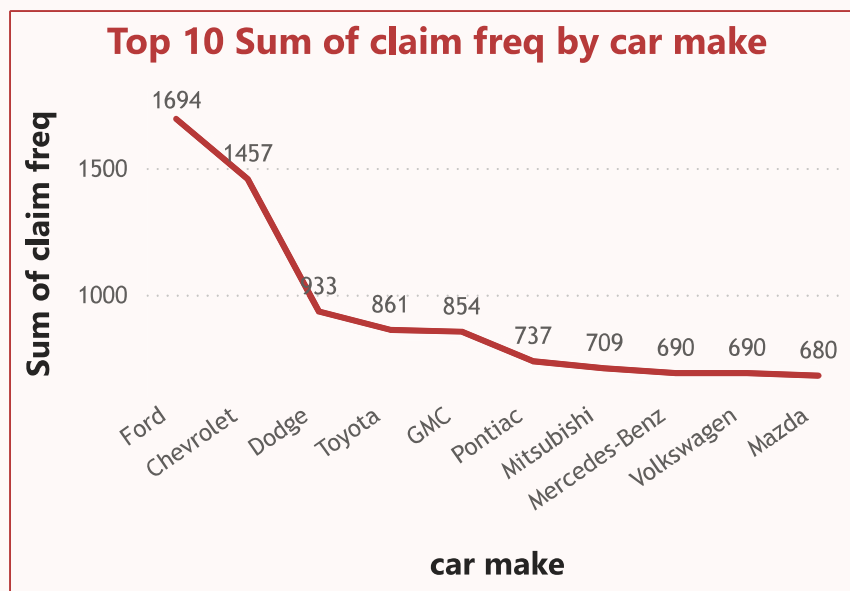
Average claim frequency by Marital Status



Average claim frequency by Gender



CLAIM FREQUENCY AND CLAIM AMOUNT ANALYSIS



coverage_zone

Highly Rural

Highly Urban

Rural

Suburban

Urban

car use

All

CLAIM FREQ AND AMOUNT

The average claim frequencies for

- **Age group:** Under 25 is 0.54
25 - 34 is 0.51
35 - 44 is 0.51
45 - 54 is 0.51
55 - 64 is 0.50
64+ is 0.51
- **Gender:** Male and Female are 0.51
- **Married Status:** Single is 0.50
Married is 0.52
Divorced is 0.49
Seperated is 0.53

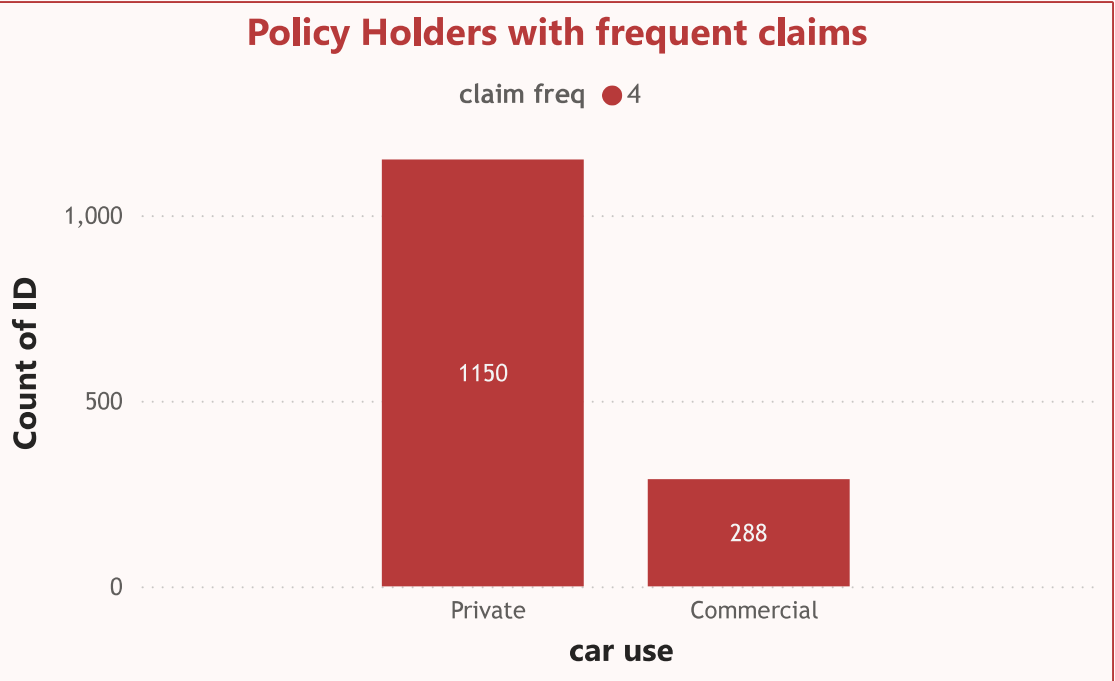
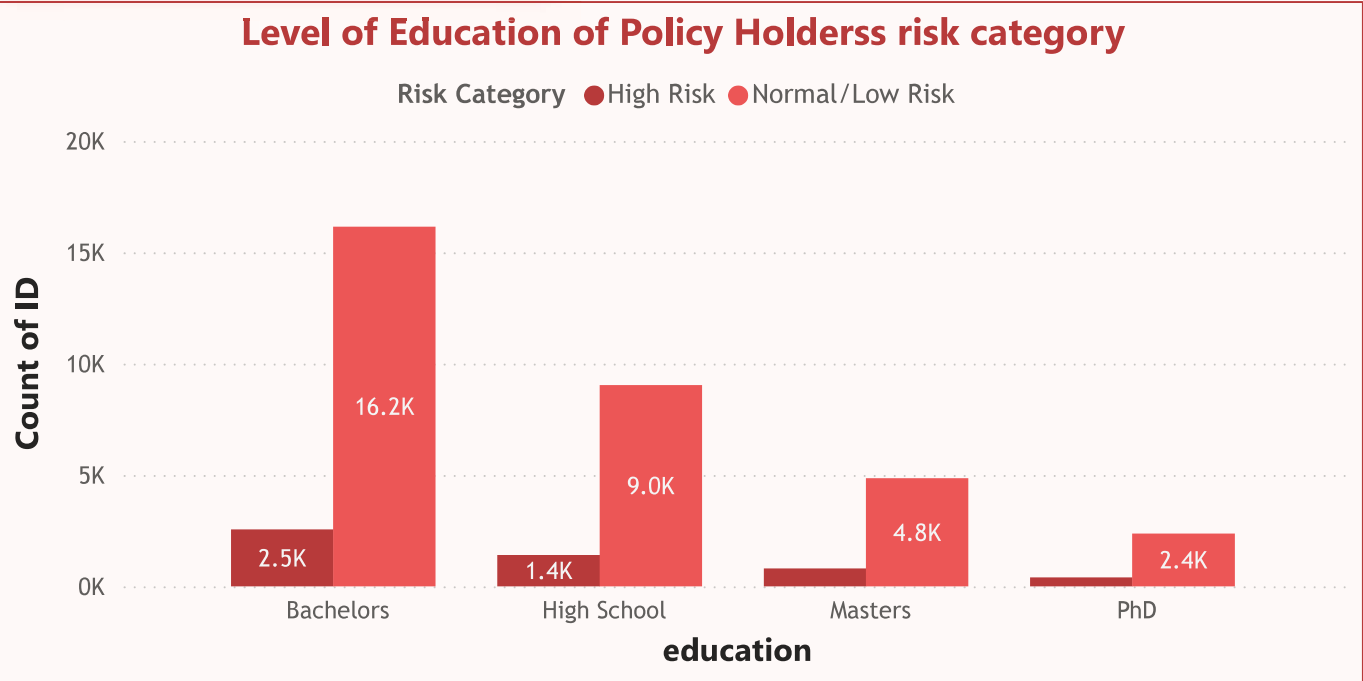
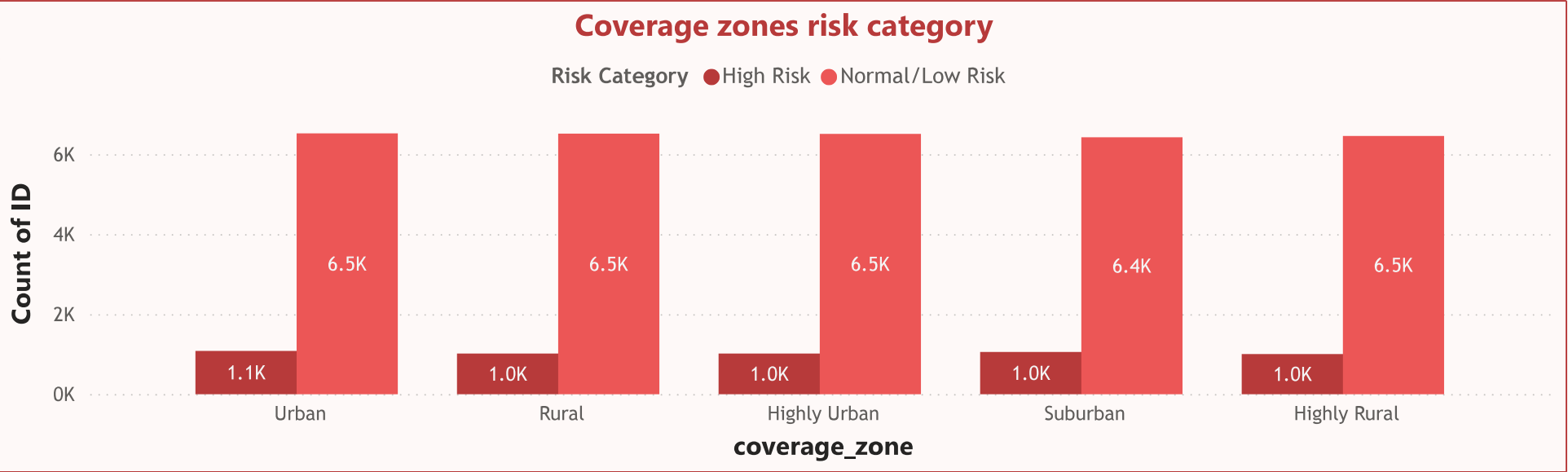
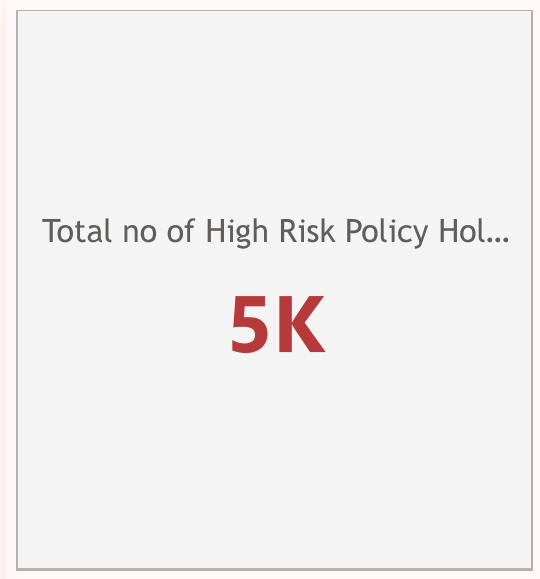
The average claim amount for

- **Age group:** Under 25 is 51k USD
25 - 34 is 50k USD
35 - 44 is 50K USD
45 - 54 is 50K USD
55 - 64 is 50K USD
64+ is 50K USD
- **Gender:** Male and Female are both 50k USD
- **Married Status:** Single is 50K USD
Married is 50K USD
Divorced is 50K USD
Seperated is 51K USD

The year 2006 correlated with the highest claim frequency and amount.

Grand Prix was the Car Model with the highest claim frequency and amount.

The car make with the highest claim frequency and amount is Ford.



POLICY HOLDERS RISK CATEGORY ANALYSIS

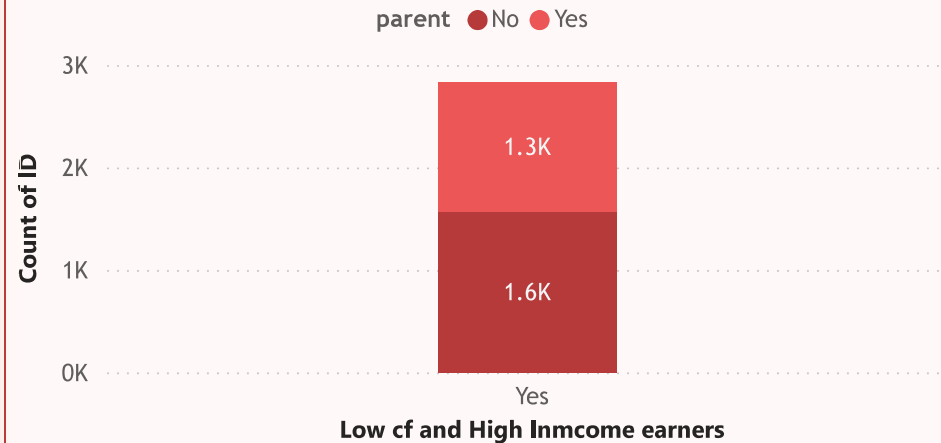
RISK ASSESSMENT

- Policy holders with a Bachelor's Degree were correlated with the highest Risk.
- Most policy holders with frequent claims were individuals who make use of their vehicles for Private use.

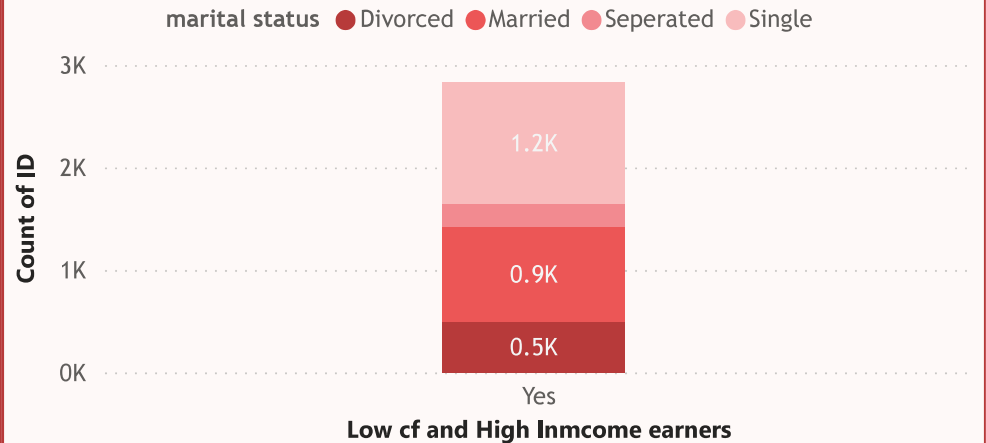
Total_no_of_policy Holders

38K

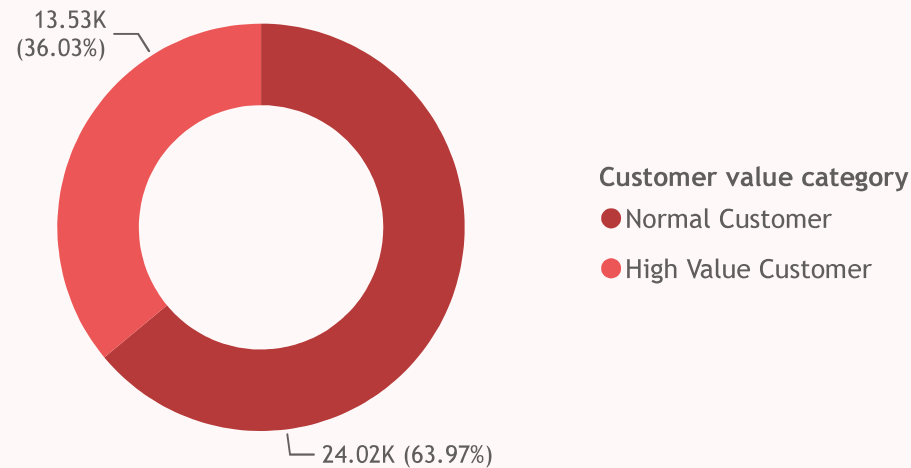
Count of ID by Low cf and High Inmcome earners and parent



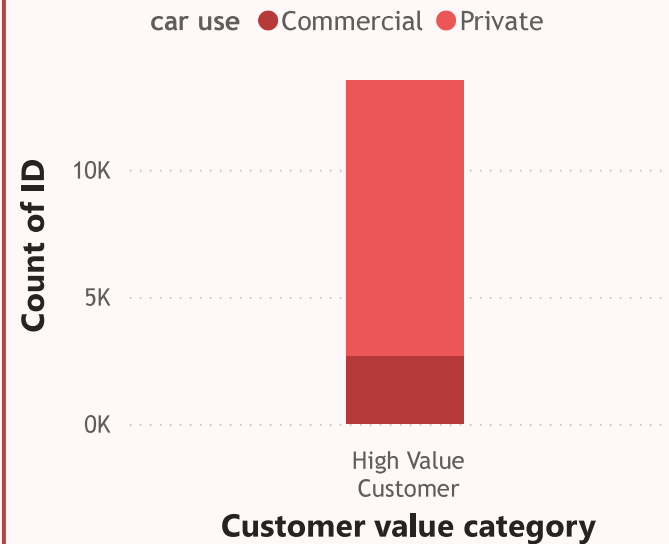
Count of ID by Low cf and High Inmcome earners and marital status



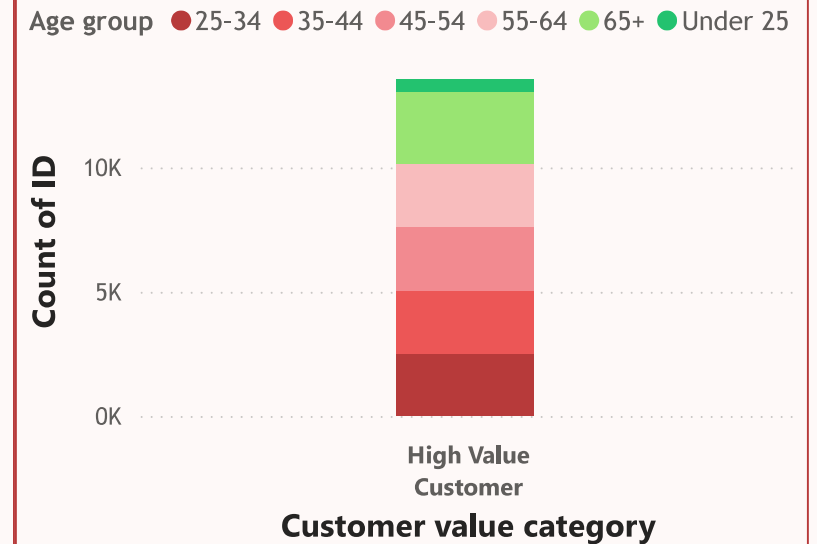
Count of ID by Customer value category



Count of ID by Customer value category and car use



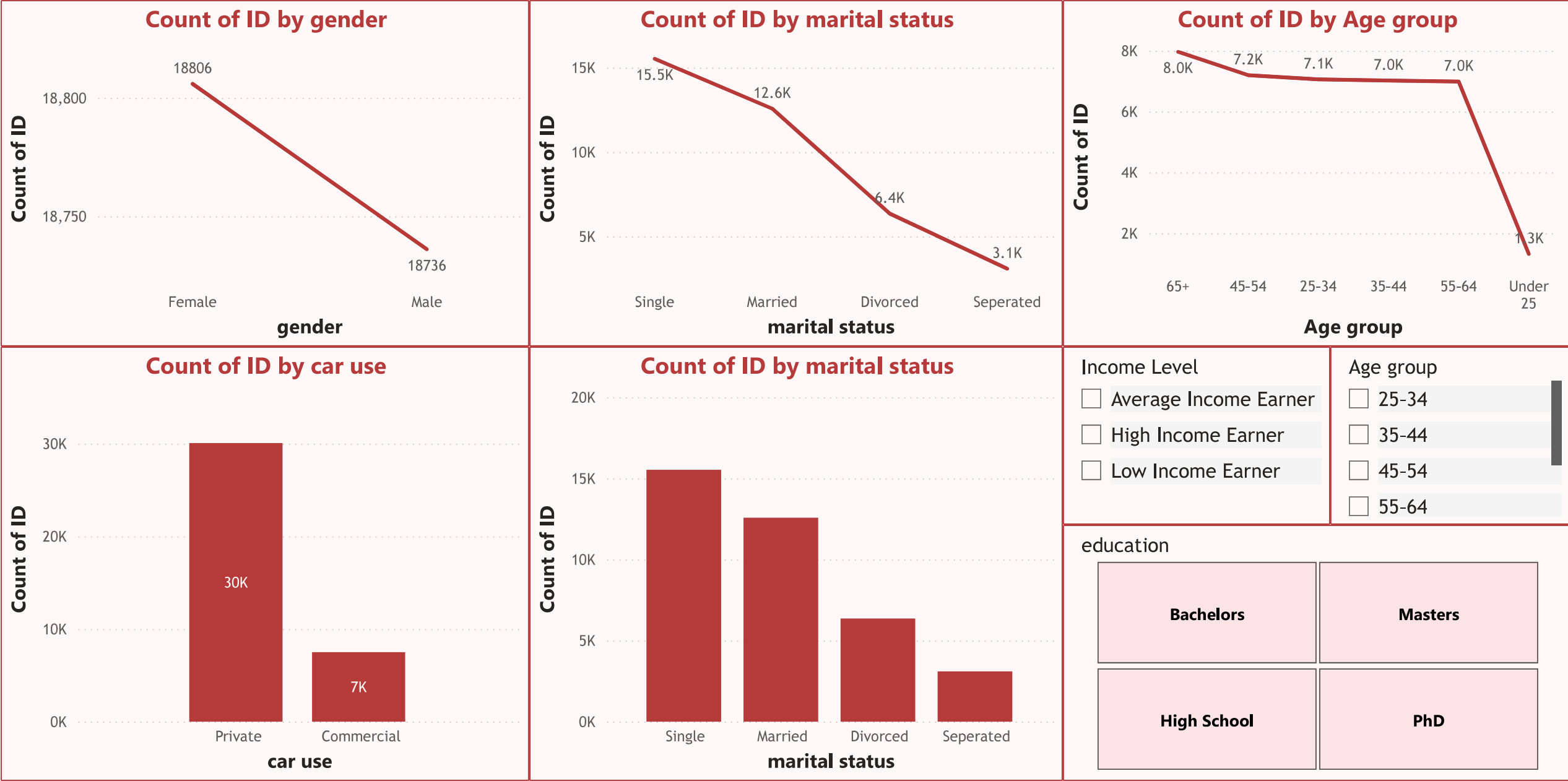
Count of ID by Customer value category and Age group



INCOME LEVEL AND CUSTOMER VALUE ANALYSIS

CUSTOMER SEGMENTAOTION

- Most policyholders with low claim frequencies and high household incomes are parents.
- Every policyholder with an household income greater than average and possessing zero(0) claim frequency was tagged a high value customer.



CUSTOMER SEGMENTATION ANALYSIS

CUSTOMER SEGMENTATION

- **Age group:**

Under 25 is 1.3k

25 - 34 is 7.1k

35 - 44 is 7.0k

45 - 54 is 7.2k

55 - 64 is 7.0k

64+ is 8.0k

- **Gender:** Male and Female are 18,736 & 18,806 respectively.

- **Marital Status:**

Singles are 15,525 in total

Married are 12,570 in total

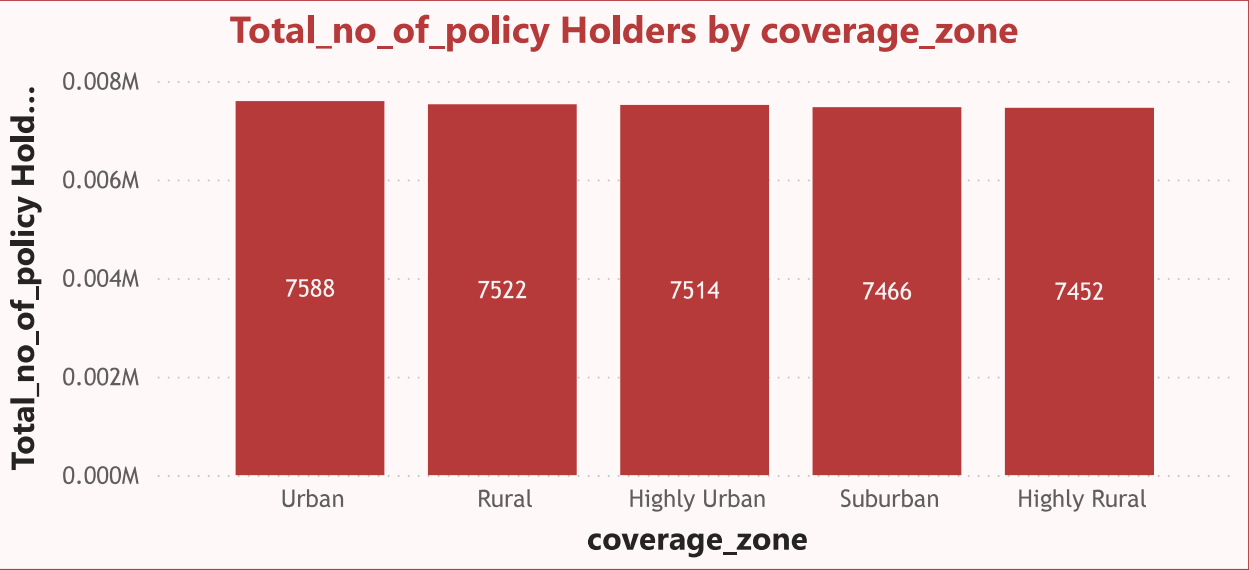
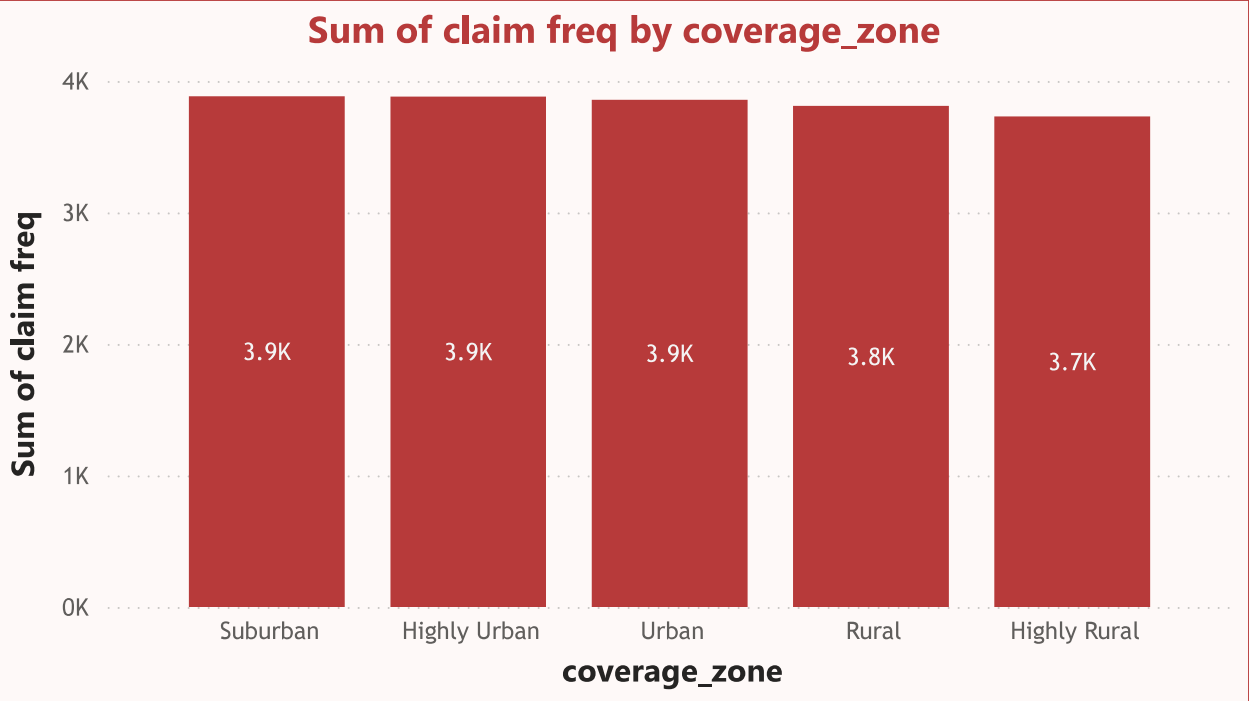
Divorced are 6,357 in total

Seperated couples are 3,090 in total

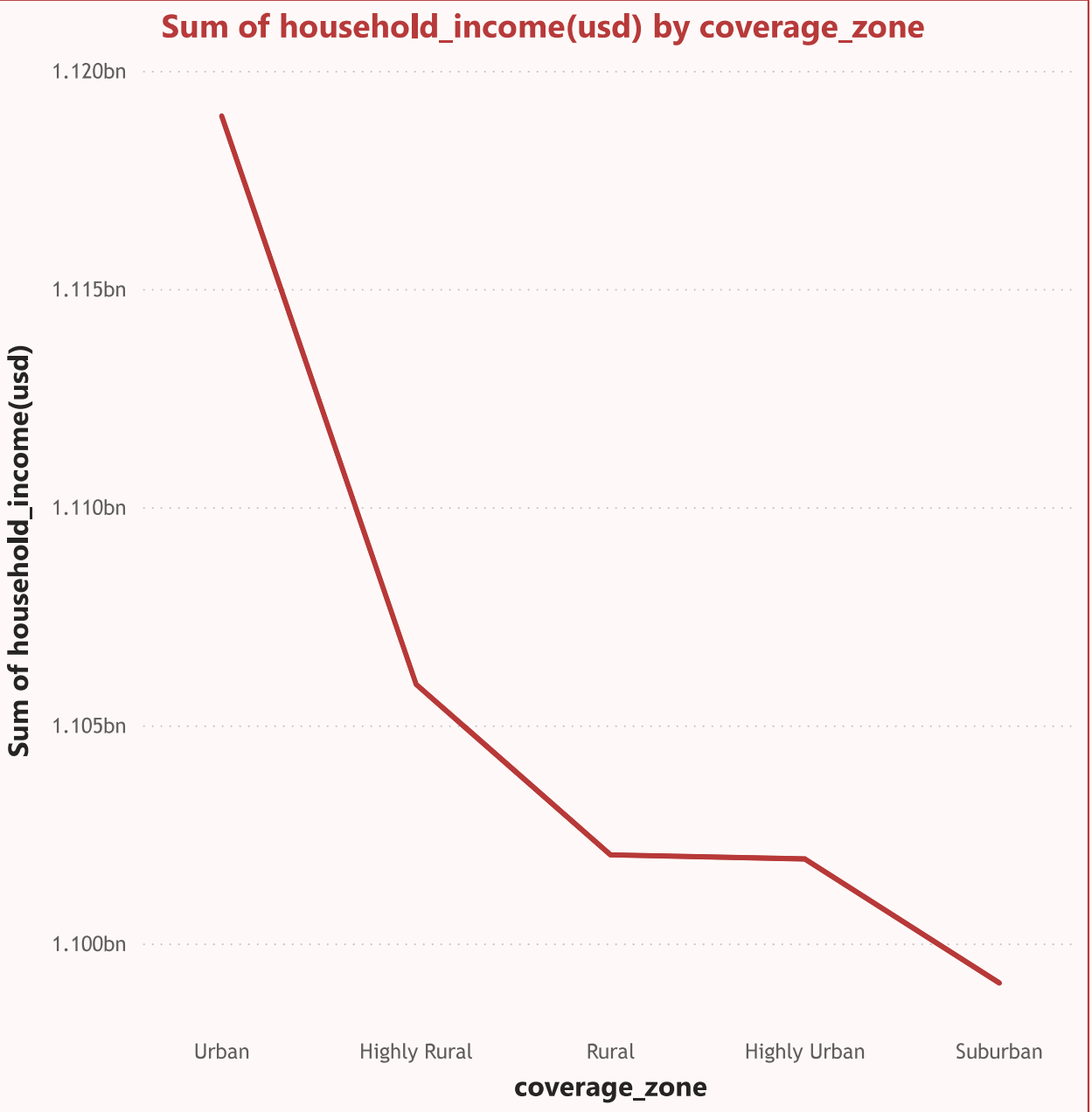
Note:

- Policy holders with private car use were more than that of commercial and single policy holders also were the highest occurred.

- They were more females than males and policyholders who fall under the 65+ age bracket were the highest in population.



How do claim frequencies and amounts vary across different coverage zones?



marital status

All

▼

DEMOGRAPHIC ANALYSIS

- Claim frequency according to Coverage Zones:

Sub Urban is 3,883

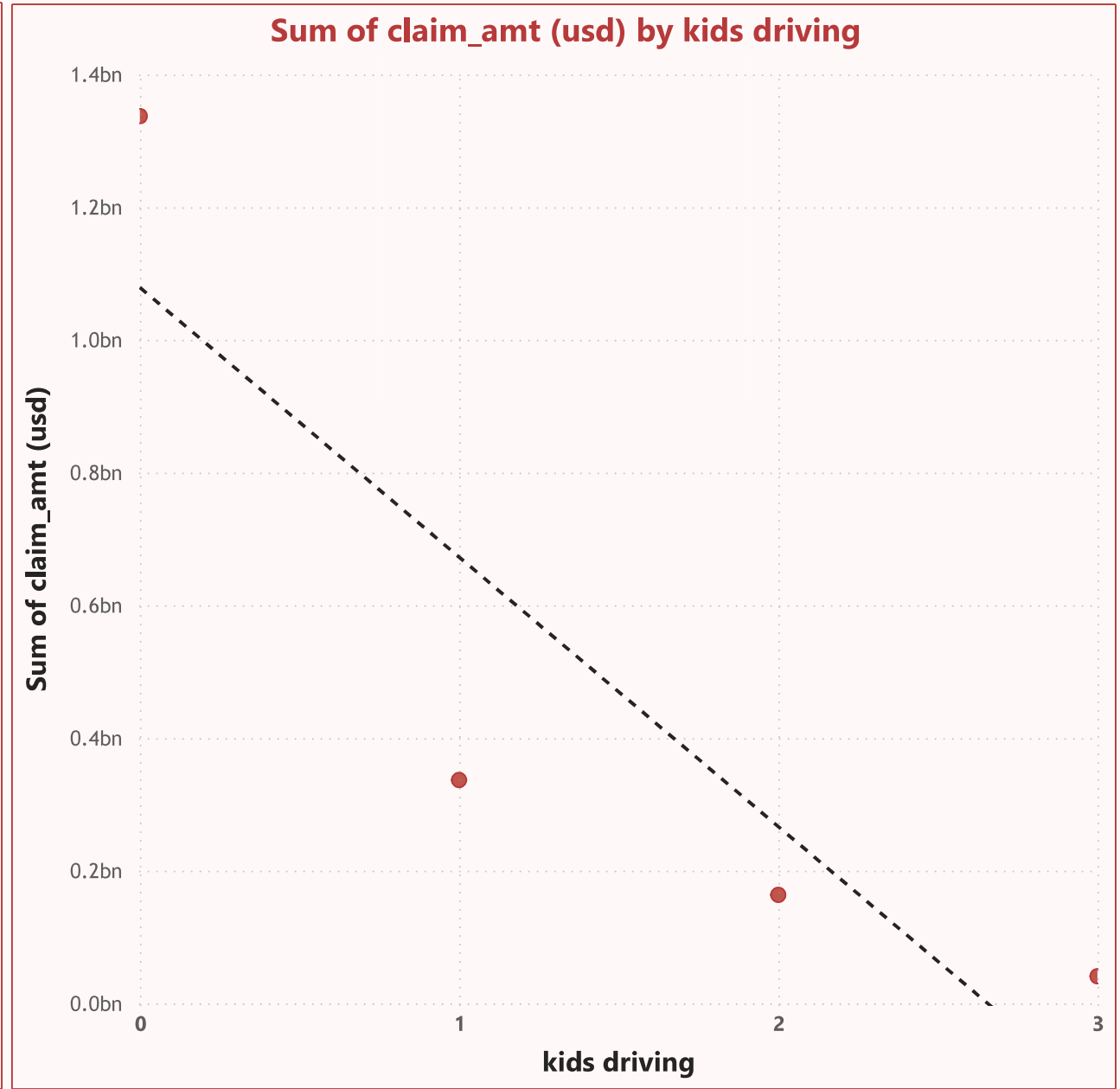
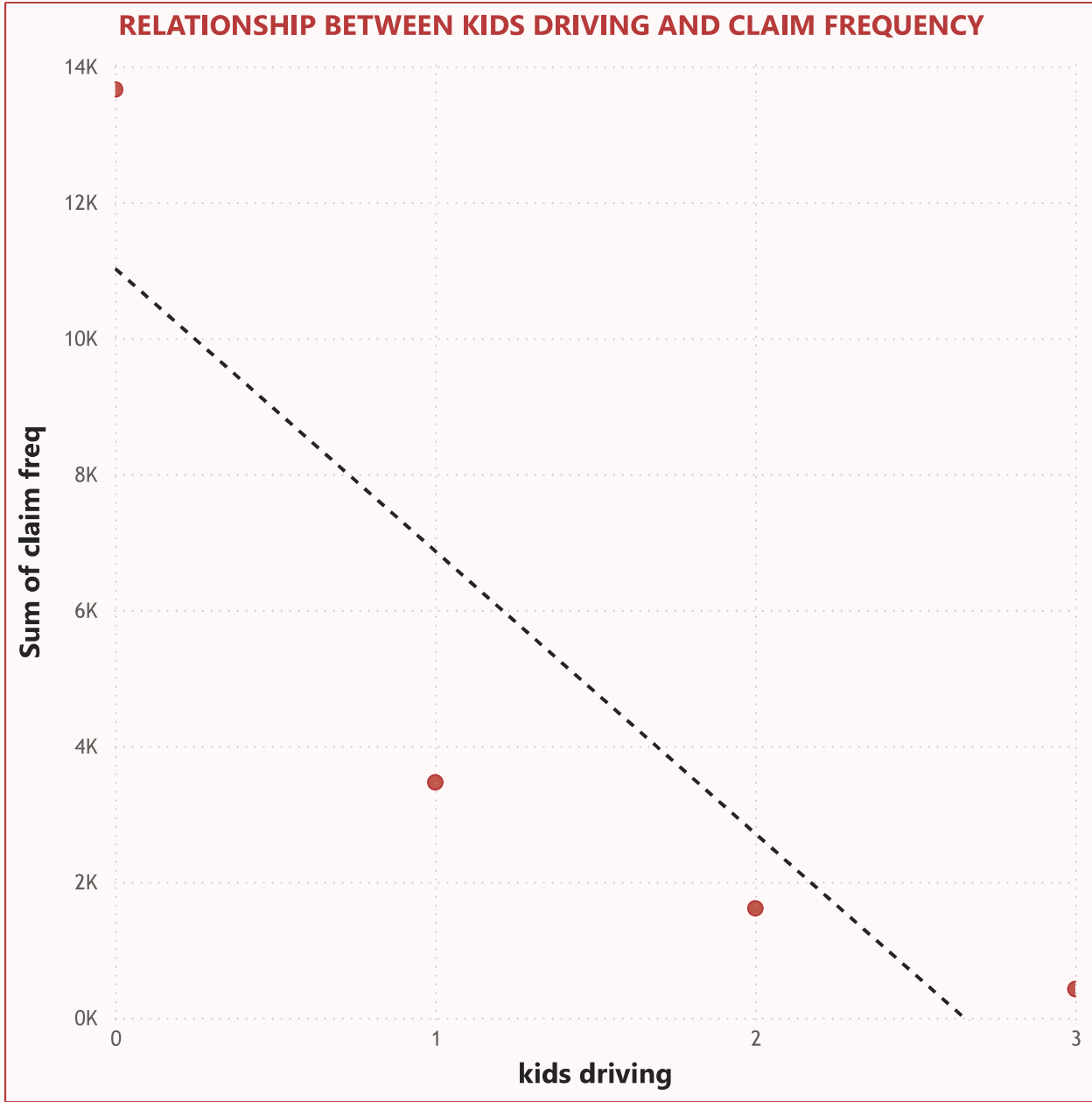
Highly Urban is 3,881

Urban is 3,856

Rural is 3,809

Highly Rural is 3,729

Note: In all regions there were relatively same amount of policyholders



CHILDREN DIRVING EFFECT ON CLAIM FREQUENCY AND AMOUNT

CUSTOMER BEHAVIOUR

- The relationship between children driving and claims were negatively correlated. As the amount of kids driving increases, claim frequency and amount reduces.

JUSTIFICATION

From a claim frequency perspective, the chart shows that households without any driving children experience substantially higher claim activity compared to households with one or more children driving. The reduction in claim frequency becomes more pronounced as additional children become licensed drivers. By the time households reach the highest category of kids driving, claim frequency falls to its lowest observed level. This pattern challenges the common assumption that the presence of more young drivers inherently increases accident risk. Instead, the data suggest that households with multiple driving children are, on average, less prone to filing insurance claims. A similar pattern emerges when claim severity is examined through total claim amount. The highest aggregate claim costs are associated with households where no children are driving, while households with one, two, or three children driving show steadily declining claim amounts. The consistency of this decline across increasing categories of kids driving provides strong

evidence that the observed relationship is robust across both frequency and severity dimensions. This dual confirmation strengthens the credibility of the findings and reduces the likelihood that the pattern is driven by data anomalies or outliers. The observed inverse relationship can be logically defended through several behavioral, economic, and risk management considerations. Households with multiple driving children are often characterized by higher levels of financial stability, better access to safer and newer vehicles, and stronger engagement with formal driver education programs. These households are more likely to implement structured rules governing driving behaviour, such as restricted driving hours, parental supervision, and controlled vehicle usage. As a result, although the number of drivers increases, risky driving behaviours and accident severity tend to decrease. Additionally, families with more than one young driver are typically more aware of insurance costs and accident consequences, which encourages more cautious driving practices. Younger drivers in such households may also have limited driving exposure, such as shorter trips or supervised driving conditions, further reducing the probability and severity of accidents. These behavioural and exposure-related factors help explain why increased driving participation among children does not translate into higher insurance losses.

From a statistical standpoint, the alignment of the data points with the downward-sloping trend lines in both panels suggests a strong and meaningful relationship between kids driving and insurance outcomes. The consistency across both claim frequency and claim amount enhances the explanatory power of the chart and supports the interpretation that the relationship is not coincidental. The parallel movement of both risk indicators in the same direction strengthens the internal validity of the analysis. In conclusion, the chart convincingly demonstrates that an increase in the number of children driving within a household is associated with a reduction in both the frequency and financial severity of insurance claims. This finding contradicts conventional assumptions about youthful driving risk and highlights the importance of household characteristics, supervision, and risk management practices in shaping insurance outcomes. The results provide valuable insights for insurers, policymakers, and risk analysts by suggesting that children's driving status should be evaluated within a broader household context rather than treated as an isolated risk factor.

GENERAL KEY INSIGHTS

- Claim frequency is largely stable across age groups, averaging ~0.50–0.54, with Under 25 showing only a marginally higher frequency.
- Claim amounts are remarkably consistent (~USD 50K) across age, gender, and marital status, indicating severity is not driven by demographics.
- Separated policyholders show a slightly higher claim frequency and amount, but not at a statistically dominant level.
- Vehicle year 2006 is associated with the highest claim frequency and severity, suggesting aging vehicles contribute to higher losses.
- Grand Prix stands out as the highest-risk model, while Ford is the highest-risk make, likely due to a combination of exposure, vehicle age, and repair costs.

According to the data and analysis;

“Risk is less about who the driver is, and more about what they drive and how old it is.”

“Uniform claim severity across demographics suggests pricing models overly dependent on age/gender may underperform.”

RECOMMENDATIONS

1. Refine Vehicle-Based Pricing Models:

- Apply risk loadings to older vehicle cohorts (especially pre-2010)
- Introduce model- and make-specific rating factors (e.g., Ford, Grand Prix).
- Vehicle Upgrade Incentives
- Offer premium discounts for newer vehicles or safety-feature-equipped cars.

Expected Effectiveness Timeline:

Short term (3–6 months): Improved pricing accuracy

Medium term (6–12 months): Reduction in claim severity from portfolio shift

2. Incorporate Usage-Based Risk Indicators:

- Differentiate private-use drivers by mileage bands, usage intensity, or driving patterns.
- Adopt Behavioral Scoring

Expected Effectiveness Timeline:

Medium term (6–9 months): Better identification of loss-prone customers

Long term (12–18 months): Portfolio loss ratio improvement

3. Create a “High-Value Customer” Segment:

- Offer loyalty rewards, bundled products, and renewal incentives.
- Targeted Upselling
- Cross-sell home, life, or umbrella insurance to this segment.

Expected Effectiveness Timeline:

Short term (3–6 months): Higher retention rates

Medium term (6–12 months): Increased customer lifetime value (CLV)

4. Tailor Products for Older Drivers:

- Emphasize safety, reliability, and service quality.
- Gender-Neutral Messaging
- Since claim outcomes are similar, avoid unnecessary gender differentiation.

Expected Effectiveness Timeline:

Short term (3–6 months): Improved engagement

Medium term (6–12 months): Better product-market fit

5. Maintain Zone-Based Pricing with Minor Adjustments:

- Apply modest urban risk loadings rather than aggressive repricing.
- Enhance Urban Risk Controls
- Promote theft protection, parking safety discounts.